TWO INVARIANTS OF AN EFFECT MONAD: CODENSITY GOVERNS FULLNESS, POLYNOMIALITY GOVERNS COMPOSITION

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Abstract. Fix a monad M on **Set**. An M-*container* is an ordinary container (S, P) whose positions are read out in the Kleisli category of M; its extension is the endofunctor $[\![S, P]\!]_M(X) = \sum_{s \in S} \mathrm{Kl}(M)(P_s, X)$, the effectful reading of a data type whose observations may fail, branch, consult a context, or return a distribution. We prove three things about the extension functor $[\![-]\!]_M \colon M\text{-}\mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$. First, M-containers are exactly $\mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}})$ and the extension factors as $[\![-]\!]_M = [\![-]\!] \circ M$: an M-container functor is an ordinary polynomial functor post-composed with the effect monad. Second, $[\![-]\!]_M$ is *always faithful*, and it is *full* if and only if M is *codense* — the canonical map $M \Rightarrow \mathrm{Ran}_M M$ into its own codensity monad is an isomorphism — and each functor $\mathbf{Set}(A, M-)$ is connected; consequently probability distributions keep fullness while error, partiality, read-only context and nondeterminism destroy it. Third, the essential image of $[\![-]\!]_M$ is closed under composition if and only if M is a *polynomial* functor — sufficiency for all M, with $p \triangleleft_M q = p \triangleleft \lceil M \rceil \triangleleft q$, and necessity for every monad that arises, via a trichotomy: super-polynomial growth is killed by a cardinality law; every analytic monad with $M\emptyset = \emptyset$ is killed by *right-cancellation of plethysm* in the ring of cycle-index series (with no bound on degree or nesting depth, so free magmas and free operad monads are covered); and every bounded-degree analytic monad is killed by a support-splitting stabilizer invariant. The only monads left uncovered are analytic with $M\emptyset$ nonempty-finite *and* unbounded degree *and* unbounded nesting depth — a corner we argue is vacuous. The affine-plus-commutative folklore criterion is false (D is affine and commutative but does not compose; Maybe is neither yet does). Fullness and composition are thus governed by two independent Kan-theoretic invariants of M that trade oppositely: probability is faithful but not composable, error is composable but not faithful.

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Draft — work in progress; not yet neighbour-reviewed. This revision supersedes the version sent to Rick on 5 September (commit `b94bc32`): Theorem 1.3's necessity direction now closes the analytic converse via plethysm right-cancellation (§5.6), retiring the previously open Plethysm Lemma.

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1. Introduction

Containers, and what happens when the positions become effectful. A *container* (S, P) — a set S of shapes together with a set P_s of positions over each shape — presents a datatype through its extension, the endofunctor

$$[\![S, P]\!](X) \;=\; \sum_{s \in S} \mathbf{Set}(P_s, X) \;=\; \sum_{s \in S} X^{P_s},$$

which sends X to "a shape together with an X-labelling of its positions" [3, 4]. The extensions of containers are exactly the *polynomial functors*, and $[\![-]\!]$ is fully faithful: a natural transformation between two such functors is precisely a container morphism. This tight correspondence — a syntax of shapes and positions on one side, a class of functors on the other — is what makes containers a usable calculus for datatypes.

Now suppose the positions are not read out purely but *through an effect.* Reading a position might fail, or return several answers, or consult an environment, or produce a probability distribution over answers. All of these are captured by a monad M on **Set**: an effectful read of a P_s-position landscape into X is a Kleisli map $P_s \to X$ in $\mathrm{Kl}(M)$, i.e. a function $P_s \to MX$. This yields the M*-container* extension

$$[\![S, P]\!]_M(X) \;=\; \sum_{s \in S} \mathrm{Kl}(M)(P_s, X) \;=\; \sum_{s \in S} (MX)^{P_s}.$$

The passage **Cont** $\to$ M-**Cont** that reinterprets a container's positions in $\mathrm{Kl}(M)$ is a change of base of exactly the kind one wants for modelling effectful data structures and effectful agents: an M-container is a device that, presented with a request, chooses a shape and then reads its positions with the ambient effect. When $M = \mathsf{Maybe}$, an "agent may decline" a position; when $M = \mathsf{D}$ (finite distributions), it answers probabilistically.

Two questions, and the temptation to conflate them. For the pure calculus, the extension functor is fully faithful and strong monoidal: it names functors faithfully, and those functors compose. Two properties, one embedding. When we move to M-containers, both properties become genuine questions:

- (Q1) **Faithful naming.** Is $[\![-]\!]_M \colon M\text{-}\mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ full and faithful? That is, does every natural transformation between M-container functors come from a (unique) M-container morphism?
- (Q2) **Composition.** Is the essential image of $[\![-]\!]_M$ closed under functor composition? That is, if F and G are M-container functors, is $F \circ G$ one too, presented by some container built from theirs?

It is tempting to expect that these rise and fall together — that "M-containers behave well" is a single condition on M. The main discovery of this note is that (Q1) and (Q2) are controlled by *two different* invariants of M, and that across the standard effects they trade against each other.

Results. Throughout, $M = (M, \eta, \mu)$ is a monad on $\mathbf{Set}$, $\mathrm{Kl}(M)$ its Kleisli category, and $[\![-]\!] \colon \mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ the ordinary (polynomial) extension.

The first theorem identifies the players and provides the lever for everything else.

Theorem 1.1 (Identification and factorisation; §3)**.** *The category of M-containers is $\mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}})$: an object is an ordinary container $(S, P \colon S \to \mathbf{Set})$, and a morphism $(S, P) \to (S', P')$ is a shape map $f \colon S \to S'$ together with, for each s, a Kleisli map $\rho_s \in \mathrm{Kl}(M)(P'_{fs}, P_s) = \mathbf{Set}(P'_{fs}, MP_s)$. Moreover, as functors $\mathbf{Set} \to \mathbf{Set}$ and naturally in (S, P),*

$$[\![S, P]\!]_M \;=\; [\![S, P]\!] \circ M.$$

So an M-container functor is an *ordinary polynomial functor pre-composed with the effect monad M*. Neil Ghani's "change of base" $\mathbf{Cont} \to M\text{-}\mathbf{Cont}$ is precomposition with the Kleisli inclusion $F \colon \mathbf{Set} \to \mathrm{Kl}(M)$, and every structural question about $[\![-]\!]_M$ becomes a question about the single trailing M.

The second theorem answers (Q1). Recall that the *codensity monad* of a functor $M \colon \mathbf{Set} \to \mathbf{Set}$ is the right Kan extension $\mathrm{Ran}_M M$; there is a canonical monad map $\iota_M \colon M \Rightarrow \mathrm{Ran}_M M$, and M is called *codense* when ι_M is an isomorphism (Definition 2.1). Call M *affine* if $M1 \cong 1$.

Theorem 1.2 (Faithfulness always; fullness = codensity; §4)**.** *The extension $[\![-]\!]_M \colon M\text{-}\mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ is* always faithful*. It is* full *— hence fully faithful — if and only if*

(i) *each functor $\mathbf{Set}(A, M-)$ is connected, and*
(ii) *M is codense.*

If M is affine then (i) *holds; hence M* affine and codense $\Rightarrow$ fully faithful*. The finite distribution monad D is affine and codense, so $[\![[\!]_M\mathsf{D}]-$ is fully faithful (modulo the rigidity Lemma 4.4); whereas* Maybe*, exceptions, the writer $E \times (-)$, the reader $(-)^R$ and nonempty powerset all fail codensity and so lose fullness.*

The correct invariant is the codensity monad, *not* preservation of coproducts. The natural first guess — "$[\![-]\!]_M$ is fully faithful iff M preserves coproducts" — is wrong for the target $[\mathbf{Set}, \mathbf{Set}]$: it is the criterion for a coarser, Kleisli-enriched target, and it gives the wrong verdict on D. The writer monad $E \times (-)$ preserves coproducts yet fails fullness for $|E| \geq 2$, a clean refutation (§4.3).

The third theorem answers (Q2). Call M *polynomial* if $M \cong \sum_{i \in I} (-)^{B_i}$ for some family $(B_i)_{i \in I}$; equivalently $M = [\![\lceil M \rceil]\!]$ for a container $\lceil M \rceil$.

Theorem 1.3 (Composition = polynomiality; §5)**.** (Sufficiency.) *If M is polynomial, then for all M-containers p, q the ordinary container*

$$p \lhd_M q \;:=\; p \lhd \lceil M \rceil \lhd q$$

satisfies $[\![p \lhd_M q]\!]_M \cong [\![p]\!]_M \circ [\![q]\!]_M$*, naturally in* X*. The operation* $\lhd_M$ *is associative and has a unit up to ordinary container isomorphism only when* $M = \mathrm{Id}$*; so the essential image of* $[\![-]\!]_M$ *is closed under composition.* (Necessity.) *Conversely, if the essential image is closed under composition then* M *is polynomial — proved for every monad that arises, by a trichotomy. Either* M *grows super-polynomially, and* $|M(MX)|$ *cannot agree on the values of* $|MX|$ *with a fixed polynomial (excluding powerset, nonempty powerset, distributions, ultrafilter, continuation); or* M *is analytic with* $M\emptyset = \emptyset$*, and* right-cancellation of plethysm *in the cycle-index ring forces* M *polynomial with no bound on degree or nesting depth (Theorem 5.14, covering free magmas and free operad monads); or* M *is analytic of bounded degree, and a support-splitting stabilizer invariant separates* $M \circ M$ *from* $[\![r]\!] \circ M$ *(Theorem 5.15, subsuming the multiset and symmetric-power monads). The only monads uncovered are analytic with* $M\emptyset$ *nonempty-finite, unbounded degree and unbounded nesting depth, a corner argued vacuous (Conjecture 5.17); modulo that the theorem is an unconditional biconditional* (*Theorem 5.18*)*.*

Combining the two theorems gives the punchline.

Corollary 1.4 (Independence and trade-off)**.** *Fullness* (codensity) *and composability* (polynomiality) *are independent invariants of* M*. The writer monad* $E \times (-)$ *is polynomial, hence composes, but is not codense, hence not full. The distribution monad* D *is affine and codense, hence full, but is not polynomial, hence does not compose.* Probability keeps faithful naming but loses composition; error keeps composition but loses faithful naming.

Method. Everything runs off the factorisation $[\![-]\!]_M = [\![-]\!] \circ M$ of Theorem 1.1. For faithfulness, the Kleisli unit η is a section, so container data can be read back off the induced natural transformation. For fullness, precomposition $(-) \circ M$ has a right adjoint Ran_M, and every hom-set $\mathrm{Nat}(\mathbf{Set}(A, M-), H)$ is a value of a right Kan extension along M; the fullness obstruction is exactly the failure of the unit $M \Rightarrow \mathrm{Ran}_M M$ to be invertible. For composition, two applications of the factorisation put every composite $[\![p]\!]_M \circ [\![q]\!]_M$ in the form $G \circ M$ with $G = [\![p]\!] \circ M \circ [\![q]\!]$; this is again an M-extension precisely when G is polynomial, which for all p, q forces M itself to be polynomial (for every monad that arises; §5.5–5.6), and then AAG strong monoidality delivers the splice $p \lhd \lceil M \rceil \lhd q$. The trailing M cannot be cancelled at the level of connected-limit preservation — Proposition 5.10 shows the single self-composite $M \circ M \cong [\![r]\!] \circ M$ carries no such information beyond the retract reduction Lemma 5.9. The converse therefore uses a *finer, faithful* invariant: an $\mathrm{Aut}(X)$-stabilizer statistic (Lemma 5.11) for bounded degree, and, for analytic M with $M\emptyset = \emptyset$, right-cancellation of plethysm on the cycle-index series (Lemma 5.13) — a genuine cancellation of the trailing M, now on the level of species isomorphism rather than limit preservation, reaching the unbounded-nesting monads no stabilizer count can touch.

Why it matters. Beyond the pleasing tension of Corollary 1.4, the result sharpens the compositional picture of effectful agents. Modelling "an agent may decline" as a Maybe-container, one keeps composition (agents chain into agents) at the price of faithfulness (distinct declining strategies can induce the same functor). Modelling "an agent answers probabilistically" as a D-container, one keeps faithful naming (strategies are recoverable from behaviour) but loses strict composition (a chain of probabilistic agents is not itself presented by a single probabilistic container — the marginal-to-joint gap). The invariant that decides each is now explicit and computable: codensity for the first question, polynomiality for the second.

Provenance and honesty. Theorems 1.1 and 1.2 (faithfulness, the codensity criterion, the affine lemma, the examples) are proved. Theorem 1.3 is proved in its sufficiency direction, and in its necessity direction for every monad that arises: the cardinality law disposes of the super-polynomial monads; plethysm right-cancellation (Lemma 5.13, Theorem 5.14) disposes of every analytic monad with $M\emptyset = \emptyset$, degree and nesting depth unbounded; and the support-splitting stabilizer invariant (Lemma 5.11, Theorems 5.12 and 5.15) disposes of every bounded-degree analytic monad. The cancellation engine (Lemma 5.13) is elementary and computationally verified; Theorems 5.14 and 5.15 additionally invoke two standard facts of combinatorial-species theory — Joyal's full faithfulness of the species-to-analytic-functor embedding, and the identity of species substitution with plethysm of cycle-index series (Joyal; Bergeron–Labelle–Leroux) — which we name as background rather than re-derive. The single class of monads not covered (analytic, $M\emptyset$ nonempty-finite, unbounded degree and nesting depth) is argued *vacuous* in Conjecture 5.17; that vacuity is the one statement we leave as a conjecture, marked where it occurs. Full faithfulness for D rests on a standard rigidity fact, $\mathrm{Nat}(\mathsf{D}, \mathsf{D}) = \{\mathrm{id}\}$, which we isolate as Lemma 4.4 and assume rather than reprove. These boundaries are marked at each occurrence.

Outline. Section 2 fixes notation: monads and Kleisli categories, the container extension and its composition monoidal structure, families over an opposite category, and right Kan extensions and codensity monads. Section 3 proves Theorem 1.1 and the always-faithfulness. Section 4 proves the codensity criterion (Theorem 1.2), works the examples, and explains why coproduct preservation is the wrong criterion. Section 5 proves Theorem 1.3: sufficiency, the cardinality law, the stabilizer invariant, and the plethysm right-cancellation that closes the analytic converse, leaving only a corner argued vacuous. Section 6 draws the trade-off and lists open problems.

2. Preliminaries

We work over **Set**. All the constructions below have counterparts over a general base along the lines of $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ [2], but the codensity and polynomiality phenomena are already visible, and cleanest, over **Set**.

2.1. **Monads and Kleisli categories.** A monad $M = (M, \eta, \mu)$ on **Set** has unit $\eta\colon \mathrm{Id} \Rightarrow M$ and multiplication $\mu\colon MM \Rightarrow M$. Its *Kleisli category* $\mathrm{Kl}(M)$ has sets as objects, $\mathrm{Kl}(M)(A, B) = \mathbf{Set}(A, MB)$, identity η_A, and composition $g \bullet f = \mu \circ Mg \circ f$. The free functor $F\colon \mathbf{Set} \to \mathrm{Kl}(M)$ is the identity on objects and sends $g\colon A \to B$ to $\eta_B \circ g$; it is faithful and bijective on objects, is a left adjoint (so preserves colimits), and is *not* full for nontrivial M. We call M *affine* if $M1 \cong 1$, and *commutative* if its two canonical strengths $MA \times MB \to M(A \times B)$ agree.

2.2. **Containers and the polynomial extension.** A *container* is a pair (S, P) with S a set and $P\colon S \to \mathbf{Set}$ (write P_s for the fibre). The category **Cont** has as morphisms $(S, P) \to (S', P')$ the pairs (f, φ) of a shape map $f\colon S \to S'$ and a reindexing $\varphi_s\colon P'_{fs} \to P_s$; equivalently $\mathbf{Cont} = \mathrm{Fam}(\mathbf{Set}^{\mathrm{op}})$ (see below). The *extension* is

$$[\![S, P]\!](X) = \sum_{s \in S} \mathbf{Set}(P_s, X), \qquad [\![S, P]\!](g) = \textstyle\sum_s (g \circ -).$$

The functor $[\![-]\!]\colon \mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ is fully faithful, with essential image the *polynomial functors* $\sum_{i \in I} (-)^{B_i}$ [3, §1],[4]. We say M is *polynomial* when M lies in this image, and write $\lceil M \rceil$ for a container with $[\![\lceil M \rceil]\!] = M$.

Cont carries the *composition* monoidal structure $(\mathbf{Cont}, \triangleleft, \mathsf{y})$, where $\mathsf{y} = (1, 1 \mapsto 1)$ is the identity container ($[\![\mathsf{y}]\!] = \mathrm{Id}$) and $\triangleleft$ is defined so that $[\![-]\!]$ is strong monoidal:

$$[\![c \triangleleft d]\!] \;\cong\; [\![c]\!] \circ [\![d]\!], \qquad [\![\mathsf{y}]\!] = \mathrm{Id}, \tag{1}$$

naturally, and $\triangleleft$ is associative up to coherent isomorphism [3, Thm. 1.12],[4]. Directed containers — containers whose $\triangleleft$-structure is a comonoid — are the categories internal to this calculus [1].

2.3. **Families over an opposite category.** For a category $\mathcal{C}$, the category $\mathrm{Fam}(\mathcal{C})$ has objects $(S, X\colon S \to \mathrm{Ob}\,\mathcal{C})$ and morphisms $(S, X) \to (S', X')$ given by a function $f\colon S \to S'$ and a family of $\mathcal{C}$-maps $X_s \to X'_{fs}$. Thus $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ has the same objects but morphisms whose component maps run $X'_{fs} \to X_s$ in $\mathcal{C}$. In particular $\mathbf{Cont} = \mathrm{Fam}(\mathbf{Set}^{\mathrm{op}})$.

2.4. **Right Kan extensions and the codensity monad.** Precomposition $M^* = (-) \circ M\colon [\mathbf{Set}, \mathbf{Set}] \to [\mathbf{Set}, \mathbf{Set}]$ has a right adjoint, the right Kan extension $\mathrm{Ran}_M\colon [\mathbf{Set}, \mathbf{Set}] \to [\mathbf{Set}, \mathbf{Set}]$ along M. The defining adjunction gives, for $H\colon \mathbf{Set} \to \mathbf{Set}$ and any representable $\mathbf{Set}(A, -)$,

$$\begin{aligned} \mathrm{Nat}\big(\mathbf{Set}(A, M-),\, H\big) &= \mathrm{Nat}\big(\mathbf{Set}(A, -) \circ M,\, H\big) \\ &= \mathrm{Nat}\big(\mathbf{Set}(A, -),\, \mathrm{Ran}_M H\big) = (\mathrm{Ran}_M H)(A), \end{aligned} \tag{2}$$

using the Yoneda lemma in the last step. Applying this with $H = M$ produces the *codensity monad* of M.

Definition 2.1. The *codensity monad* of M is $\Phi := \mathrm{Ran}_M M$; by (2), $\Phi(A) = \mathrm{Nat}(\mathbf{Set}(A, M-), M)$. The unit η of the adjunction $M^* \dashv \mathrm{Ran}_M$, evaluated at M, is a natural monad map $\iota_M \colon M \Rightarrow \Phi$, explicitly $\iota_{M,A}(m) = [\, k \mapsto \mu_A(Mk\,(m))\,]$. We call M *codense* when ι_M is an isomorphism.

Definition 2.2. A functor $G\colon \mathbf{Set} \to \mathbf{Set}$ is *connected* if $\mathrm{Nat}(G, -)$ preserves coproducts: $\mathrm{Nat}(G, \sum_j H_j) = \sum_j \mathrm{Nat}(G, H_j)$ for every family (H_j). Equivalently, every natural transformation out of G into a coproduct factors through a single summand.

3. The identification, and always-faithfulness

Definition 3.1. An *M-container* is a container (S, P); its *M-extension* is

$$[\![S, P]\!]_M(X) = \sum_{s\in S} \mathrm{Kl}(M)(P_s, X) = \sum_{s\in S} \mathbf{Set}(P_s, MX),$$

with action $[\![S, P]\!]_M(g) = \sum_s (Mg \circ -)$ on a map $g\colon X \to X'$. A morphism $(S, P) \to (S', P')$ of M-containers is a shape map $f\colon S \to S'$ together with, for each s, a Kleisli map $\rho_s \in \mathrm{Kl}(M)(P'_{fs}, P_s) = \mathbf{Set}(P'_{fs}, MP_s)$; composition uses $\bullet$ in $\mathrm{Kl}(M)$.

Proof of Theorem 1.1. (a) M-$\mathbf{Cont} = \mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}})$. Objects of $\mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}})$ are pairs $(S, P\colon S \to \mathrm{Ob}\,\mathrm{Kl}(M))$, and $\mathrm{Ob}\,\mathrm{Kl}(M) = \mathbf{Set}$, so they are exactly containers. A morphism $(S, P) \to (S', P')$ is $f\colon S \to S'$ together with, per s, a $\mathrm{Kl}(M)^{\mathrm{op}}$-map $P_s \to P'_{fs}$, i.e. a $\mathrm{Kl}(M)$-map $P'_{fs} \to P_s$, i.e. a function $P'_{fs} \to MP_s$ — exactly Definition 3.1. Identities match (Fam's identity uses id in $\mathrm{Kl}(M)^{\mathrm{op}}$, i.e. η, the M-container identity), and composition matches (Fam composes shapes in $\mathbf{Set}$ and positions by $\bullet$ in the opposite order). The two categories are equal on objects, hom-sets, identities and composition.

(b) $[\![S, P]\!]_M = [\![S, P]\!] \circ M$. On objects, $[\![S, P]\!](MX) = \sum_s \mathbf{Set}(P_s, MX) = [\![S, P]\!]_M(X)$; on $g\colon X \to X'$ both sides act by $\sum_s(-\circ Mg)$. For naturality in (S, P), a morphism (f, ρ) induces $\alpha_X(s, k) = (fs,\ k\bullet\rho_s) = (fs,\ \mu_X \circ Mk \circ \rho_s)$, which is precisely $[\![-]\!] \circ M$ applied to the underlying shape/position data. ☐

Part (b) is the lever: *an M-container functor is a polynomial functor with a trailing M.*

Proposition 3.2 (Always faithful)**.** *$[\![-]\!]_M\colon M$-$\mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ is faithful.*

Proof. Suppose (f, ρ) and $(\tilde f, \tilde\rho)$ induce the same natural transformation α. Evaluate at $X = P_s$ on the element $(s, \eta_{P_s}) \in [\![S, P]\!]_M(P_s)$: $\alpha_{P_s}(s, \eta_{P_s}) = (fs,\ \eta_{P_s}\bullet\rho_s) = (fs, \rho_s)$, since η is the Kleisli identity. Equality of α therefore forces $fs = \tilde f s$ and $\rho_s = \tilde\rho_s$ for every s. Hence $(f, \rho) = (\tilde f, \tilde\rho)$. ☐

Remark 3.3. Faithfulness uses only that η is a section of the Kleisli identity; it holds for every monad. This already separates M-containers from linear (vector-space) containers, whose extension is not faithful because the analogue of η is not a section.

4. Fullness is codensity

Fix M-containers (S,P) and (S',P'). By Theorem 1.1(b), $[\![S,P]\!]_M = p\circ M$ and $[\![S',P']\!]_M = p'\circ M$ with $p=[\![S,P]\!]$, $p'=[\![S',P']\!]$. Since $p=\sum_s \mathbf{Set}(P_s,-)$ is a coproduct of representables,

$$\mathrm{Nat}(p\circ M,\ p'\circ M) = \prod_s \mathrm{Nat}\Big(\mathbf{Set}(P_s, M-),\ \sum_{s'} \mathbf{Set}(P'_{s'}, M-)\Big), \tag{3}$$

while the M-container hom-set is $\prod_s \sum_{s'} \mathbf{Set}(P'_{s'}, MP_s)$. Comparing, fullness holds for all pairs if and only if, for every set A (playing the role of P_s) and every family $(P'_{s'})$, the map

$$\begin{aligned} \sum_{s'} \mathbf{Set}(P'_{s'}, MA) \ &\longrightarrow\ \mathrm{Nat}\Big(\mathbf{Set}(A, M-),\ \sum_{s'} \mathbf{Set}(P'_{s'}, M-)\Big), \\ (s',\rho)\ &\mapsto\ [\,k\mapsto (s', k\bullet\rho)\,], \end{aligned} \tag{$\star$}$$

is a bijection.

4.1. **The Kan engine and the split.** By (2) with A, every natural transformation out of $\mathbf{Set}(A,M-)$ is a value of a right Kan extension along M. Two features of $\mathbf{Set}$ let us split ($\star$) cleanly. First, a single summand is a product of copies of M: $\mathbf{Set}(B,M-)=\prod_{b\in B} M$, so $\mathrm{Nat}(\mathbf{Set}(A,M-),\mathbf{Set}(B,M-)) = \mathrm{Nat}(\mathbf{Set}(A,M-),M)^B = \Phi(A)^B$ by Definition 2.1. Second, the coproduct over s' is handled exactly when $\mathbf{Set}(A,M-)$ is connected. This yields:

Proof of Theorem 1.2. Faithfulness is Proposition 3.2. For fullness we show ($\star$) is a bijection for all $A,(P'_{s'})$ iff (i) and (ii) hold.

(ii) handles a single summand. With one summand $B=P'_{s'}$, ($\star$) becomes $\mathbf{Set}(B,MA)\to \mathrm{Nat}(\mathbf{Set}(A,M-),\mathbf{Set}(B,M-)) = \Phi(A)^B$, which is $\big(MA\to \Phi(A)\big)^B$; and the map is $\iota_{M,A}$ in each coordinate. So the single-summand case for all B is a bijection iff $\iota_{M,A}\colon MA\to\Phi(A)$ is a bijection for all A — that is, iff M is codense.

(i) handles the coproduct. Given codensity, the general ($\star$) is a bijection iff the canonical comparison, with $G_{s'}=\mathbf{Set}(P'_{s'},M-)$,

$$\textstyle\sum_{s'} \mathrm{Nat}(\mathbf{Set}(A,M-),G_{s'}) \ \longrightarrow\ \mathrm{Nat}(\mathbf{Set}(A,M-),\sum_{s'} G_{s'})$$

is a bijection — that is, iff $\mathbf{Set}(A,M-)$ is connected.

Hence ($\star$) for all data $\iff$ (i)$\wedge$(ii), proving the criterion. $\square$

Lemma 4.1 (Affine $\Rightarrow$ connected)**.** *If M is affine then each $\mathbf{Set}(A,M-)$ is connected; hence M affine and codense $\Rightarrow [\![-]\!]_M$ is fully faithful.*

Proof. Since $M1 \cong 1$, $\mathbf{Set}(A, M1) = (M1)^A \cong 1$: the functor $\mathbf{Set}(A, M-)$ has a single global point. Let $\alpha\colon \mathbf{Set}(A, M-) \Rightarrow \sum_{s'} G_{s'}$. The unique point of $\mathbf{Set}(A, M1)$ maps to a single summand s'_0 of $\sum_{s'} G_{s'}(1)$. For any X, the terminal map $!\colon X \to 1$ gives a commuting square; since $\sum_{s'} G_{s'}(!)$ preserves summands and the image of the global point lies in s'_0, naturality forces α_X to land in the s'_0-summand for every X. Thus α factors through s'_0, which is connectedness. □

4.2. **The examples.** For a polynomial monad $M = \sum_{i\in I}(-)^{B_i}$ the whole comparison is computable by (1): $\mathbf{Set}(A, M-)$ is again polynomial, with shapes $\varphi\colon A \to I$ and positions $\sum_a B_{\varphi a}$, so that

$$|MA| = \sum_i |A|^{|B_i|}, \qquad |\Phi(A)| = \prod_{\varphi\colon A\to I} \sum_{i\in I} \Big(\textstyle\sum_a |B_{\varphi a}|\Big)^{|B_i|}.$$

Codensity is the equality $|MA| = |\Phi(A)|$ for all A. Exact computation gives:

M	$M1$	$\lvert MA\rvert$ vs $\lvert\Phi(A)\rvert$ $(A = 0, 1, 2, 3)$	codense?	full?
Id	1	$0, 1, 2, 3 \;=\; 0, 1, 2, 3$	yes	**yes**
Maybe $(X{+}1)$	2	$1, 2, 3, 4$ vs $1, 2, \mathbf{12}, \mathbf{864}$	no	no
exception $X{+}2$	3	$2, 3, 4, 5$ vs $2, \mathbf{12}, \mathbf{5184}, \ldots$	no	no
writer $2{\times}X$	2	$0, 2, 4, 6$ vs $0, \mathbf{4}, \mathbf{256}, \ldots$	no	no
reader X^2	1	$0, 1, 4, 9$ vs $0, \mathbf{4}, \mathbf{16}, \mathbf{36}$	no	no

Beyond a low threshold $|\Phi(A)| > |MA|$ strictly in every nontrivial case: the surplus natural transformations are "case-analysis" operations — inspecting $\perp$, or duplicating a reader argument — that are natural for pure maps but not Kleisli-natural.

Example 4.2 (Reader: affine but not codense)**.** The reader monad $(-)^R$ is affine ($M1 = 1$, so (i) holds by Lemma 4.1) yet fails (ii): $\Phi(A) = (RA)^R \neq A^R = MA$ for $|R| \geq 2$. Affineness alone does not give fullness; codensity is a genuinely separate necessary condition.

Example 4.3 (Distributions: affine and codense)**.** Let D be the finitely-supported distribution monad. It is affine ($\mathsf{D}1 \cong 1$). We claim it is codense: $\Phi(A) = \mathrm{Nat}(\mathsf{D}^A, \mathsf{D}) = \mathsf{D}A$, i.e. the natural A-ary operations on D are exactly the affine (convex) combinations. Given $\alpha\colon \mathsf{D}^A \Rightarrow \mathsf{D}$ and a tuple $(\mu_a)_a \in (\mathsf{D}X)^A$, put $K = \coprod_a \mathrm{supp}(\mu_a)$, let $\nu_a \in \mathsf{D}K$ be the block-a copy of μ_a, and $p\colon K \to X$ the projection; then $(\mu_a) = \mathsf{D}^A(p)((\nu_a))$, so $\alpha_X((\mu_a)) = \mathsf{D}p\,(\alpha_K((\nu_a)))$. Collapsing the blocks by $c\colon K \to A$ gives $\mathsf{D}c\,(\alpha_K((\nu_a))) = \alpha_A((\delta_a)) =: \omega \in \mathsf{D}A$. Reducing one block at a time (collapse all but one block to a point) reduces to classifying $\mathrm{Nat}(\mathsf{D}, \mathsf{D}(-\sqcup F))$ for finite F: the F-marginal is constant (it factors through $\mathsf{D}1 = 1$) and the remaining marginal is a natural sub-probability-valued endofunctor of D, hence a scalar $\lambda \cdot \sigma$. Thus $\alpha_K((\nu_a)) = \sum_a \omega_a \nu_a$ and $\alpha_X((\mu_a)) = \sum_a \omega_a \mu_a$. So D is codense, and by Lemma 4.1 $[\![ ]\!]_M\mathsf{D}]-$ *is fully faithful* — modulo the base rigidity Lemma 4.4.

Lemma 4.4 (Rigidity of D; assumed)**.** *The only natural endomorphism of* D *is the identity, and every natural sub-probability-valued endofunctor of* D *is a scalar multiple of the identity:* $\mathrm{Nat}(\mathsf{D},\mathsf{D}) = \{\mathrm{id}\}$.

This is a standard rigidity fact for the affine functor D; we assume it rather than reprove it, and we mark the full faithfulness of $[\![[\![]\!]_M \mathsf{D}]\!]-$ as conditional on it. The reduction in Example 4.3 is unconditional.

4.3. **Why coproduct preservation is the wrong criterion.** A natural first guess is that $[\![-]\!]_M$ is fully faithful iff M preserves coproducts. This is false for the target $[\mathbf{Set},\mathbf{Set}]$, for a structural reason. Factor

$$[\![-]\!]_M \;=\; R \circ [\![-]\!]^{\mathrm{Kl}}, \qquad R = (- \circ F),$$

where $[\![-]\!]^{\mathrm{Kl}}\colon \mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}}) \to [\mathrm{Kl}(M),\mathrm{Kl}(M)]$ is the free-coproduct-completion extension (always fully faithful) and R restricts along the free functor $F\colon \mathbf{Set} \to \mathrm{Kl}(M)$. "$M$ preserves coproducts" is the fullness criterion for the coarser target $[\mathrm{Kl}(M),\mathrm{Kl}(M)]$ — the connected-unit condition for $[\![-]\!]^{\mathrm{Kl}}$ — and $[\![-]\!]^{\mathrm{Kl}}$ is already fully faithful. All the genuine content is in R, i.e. in the gap between pure-naturality and Kleisli-naturality, and that gap is measured by the codensity monad, not by coproduct preservation.

The verdicts differ concretely. Coproduct preservation fails for D, so the wrong criterion predicts D fails fullness; the correct criterion (affine and codense) says D *keeps* fullness. Conversely the writer monad $E\times(-)$ *preserves* coproducts, so the wrong criterion predicts fullness, yet $\Phi(1) = |E|^{|E|} \neq |E|$ for $|E| \geq 2$, so $E \times (-)$ is not codense and $[\![-]\!]_M$ is not full. The writer monad is thus a clean refutation of the coproduct criterion.

Remark 4.5 (Corrected slogan)**.** Adding probabilistic positions preserves the on-the-nose faithful naming of the container extension; adding error, partiality, read-only context or nondeterminism destroys it. The dividing invariant is codensity of the effect monad, computed by the right Kan extension $\mathrm{Ran}_M M$.

5. Composition is polynomiality

We now turn to (Q2). Throughout, $[\![-]\!]\colon \mathbf{Cont} \to [\mathbf{Set},\mathbf{Set}]$ is the strong monoidal extension of (1).

5.1. **The structural identity.**

Lemma 5.1. *For all M-containers p, q, as functors* $\mathbf{Set} \to \mathbf{Set}$ *and naturally in X,*

$$[\![p]\!]_M \circ [\![q]\!]_M \;=\; G \circ M, \qquad G := [\![p]\!] \circ M \circ [\![q]\!].$$

Proof. Two applications of Theorem 1.1(b) and associativity of $\circ$:

$$([\![p]\!]_M \circ [\![q]\!]_M)(X) = [\![p]\!]_M\big([\![q]\!](MX)\big) = [\![p]\!]\big(M([\![q]\!](MX))\big) = ([\![p]\!]\circ M \circ [\![q]\!] \circ M)(X).$$

∎

So every composite of two M-extensions has a trailing M; it is again an M-extension exactly when the inner G is polynomial.

5.2. **Sufficiency.**

Proposition 5.2 (Sufficiency)**.** *If M is polynomial with container $\lceil M \rceil$, then $p \lhd_M q := p \lhd \lceil M \rceil \lhd q$ satisfies $[\![p \lhd_M q]\!]_M \cong [\![p]\!]_M \circ [\![q]\!]_M$ naturally in X. The operation $\lhd_M$ on* **Cont** *is associative up to the $\lhd$-associator.*

Proof. With $M = [\![\lceil M \rceil]\!]$, (1) applied twice gives

$$G = [\![p]\!] \circ M \circ [\![q]\!] = [\![p]\!] \circ [\![\lceil M \rceil]\!] \circ [\![q]\!] \cong [\![p \lhd \lceil M \rceil \lhd q]\!] = [\![r]\!], \qquad r := p \lhd \lceil M \rceil \lhd q,$$

naturally, the two bracketings agreeing by associativity of $\lhd$. Whiskering by M on the right and invoking Lemma 5.1,

$$[\![p]\!]_M \circ [\![q]\!]_M = G \circ M \cong [\![r]\!] \circ M = [\![r]\!]_M,$$

naturally in X. Associativity of $\lhd_M$ is associativity of $\lhd$: $(p \lhd_M q) \lhd_M s = p \lhd \lceil M \rceil \lhd q \lhd \lceil M \rceil \lhd s = p \lhd_M (q \lhd_M s)$. □

Proposition 5.3 (Non-unitality)**.** *The operation $\lhd_M$ has a two-sided unit up to ordinary container isomorphism if and only if $M = \mathrm{Id}$.*

Proof. If $M = \mathrm{Id}$ then $\lceil M \rceil = \mathsf{y}$ and $p \lhd_M q = p \lhd q$, with unit y. Conversely, let e be a two-sided unit up to **Cont**-isomorphism. Taking $q = \mathsf{y}$ in the left unit law, $e \lhd_M \mathsf{y} = e \lhd \lceil M \rceil \cong \mathsf{y}$; applying the fully faithful $[\![-]\!]$ and (1) gives $[\![e]\!] \circ M \cong \mathrm{Id}$. Write $[\![e]\!] = \sum_{s \in S_e} \mathbf{Set}(E_s, -)$, so $\sum_s \mathbf{Set}(E_s, M-) \cong \mathrm{Id}$. Evaluate:

- At $X = 1$: $\sum_s |M1|^{|E_s|} = 1$. Each summand is ≥ 1 (as $|M1| \geq 1$ by η_1), so there is a unique shape s_0 with $|M1|^{|E_{s_0}|} = 1$; and $E_{s_0} = \emptyset$ is impossible ($\mathbf{Set}(\emptyset, M-) = \mathrm{const}_1$ gives value $1 \neq 2$ at $X = 2$). Hence $|M1| = 1$ and $\mathbf{Set}(E_{s_0}, M-) \cong \mathrm{Id}$.
- At $X = 2$: $|M2|^{|E_{s_0}|} = 2$. If $|E_{s_0}| \geq 2$ this is 1 or ≥ 4, never 2; so $|E_{s_0}| = 1$ and $|M2| = 2$.

Thus $E_{s_0} \cong 1$ and $\mathbf{Set}(1, M-) = M \cong \mathrm{Id}$. □

Remark 5.4. The clean statement is a unit up to *ordinary container* isomorphism, obtained by pushing the unit law through the fully faithful $[\![-]\!]$. A unit up to the coarser M-**Cont** (Kleisli) isomorphism is a more delicate question we do not address. For applications the operative fact is: $(\mathbf{Cont}, \lhd_M)$ is semigroupal, and monoidal only at $M = \mathrm{Id}$.

Remark 5.5 (Scope on morphisms)**.** Proposition 5.2 defines $\lhd_M$ on objects and makes $[\![-]\!]_M$ strong semigroupal on the subcategory of *pure* morphisms (those of the form $\eta \circ g$). Extending $\lhd_M$ to a bifunctor on all of M-**Cont** so that $[\![-]\!]_M$ is strong semigroupal requires transporting horizontal composites back along $[\![-]\!]_M$, which is free exactly when $[\![-]\!]_M$ is full — i.e. when M is codense (Theorem 1.2). Thus the object-level closure holds for every polynomial M, and upgrades to the full Kleisli category precisely when M is additionally codense. This Kleisli-bifunctoriality is the exact meeting point of Theorems 1.2 and 1.3; we leave it open.

5.3. **Necessity: the cardinality growth law.** Sufficiency looks like half of a biconditional, but the converse is subtler. We prove a cardinality obstruction that already decides all the affine and commutative candidates, then isolate the one remaining categorical gap.

Proposition 5.6 (Cardinality growth law)**.** *Suppose the essential image* $\{[\![r]\!] \circ M : r \in \mathbf{Cont}\}$ *is closed under* $\circ$*. Then for every container* q *there is a fixed nonnegative-integer polynomial* R_q *with, writing* $m(n) = |M(n)|$ *and* Q *the cardinality polynomial of* $[\![q]\!]$*,*

$$m\big(Q(m(n))\big) = R_q\big(m(n)\big) \qquad \text{for all finite } n.$$

In particular, at $q = \mathsf{y}$ *(so* $Q = \mathrm{Id}$*),* $m(m(n)) = R(m(n))$*: the function* m *agrees on* $\mathrm{im}(m)$ *with a fixed nonnegative-integer polynomial.*

Proof. The identity container y has $[\![\mathsf{y}]\!]_M = M$, and every $[\![q]\!]_M = [\![q]\!] \circ M$ lies in the image. Closure applied to $p = \mathsf{y}$ and general q gives, via Lemma 5.1, $M \circ [\![q]\!] \circ M = [\![\mathsf{y}]\!]_M \circ [\![q]\!]_M \cong [\![r_q]\!] \circ M$ for some container r_q. Taking cardinalities on a finite set of size n yields $m(Q(m(n))) = R_q(m(n))$ with R_q the (fixed) cardinality polynomial of r_q. The case $q = \mathsf{y}$ is $Q = \mathrm{Id}$. □

Corollary 5.7 (The affine/commutative folklore is false)**.** *Nonempty powerset* $\mathcal{P}^+$*, powerset* $\mathcal{P}$ *and the distribution monad* D *all fail composition, already at* $p = q = \mathsf{y}$*; while* Maybe *and the writer* $E \times (-)$ *compose. In particular "composes* $\iff$ *commutative and affine" is false in both directions.*

Proof. For $\mathcal{P}^+$, $m(n) = 2^n - 1$ and $m(m(n)) = 2^{2^n-1} - 1$ grows doubly exponentially in n, hence super-polynomially in $v = m(n)$; no fixed polynomial R satisfies $R(2^n - 1) = 2^{2^n-1} - 1$ for all n, so Proposition 5.6 fails at $q = \mathsf{y}$. The same argument applies to $\mathcal{P}$ ($m(n) = 2^n$). For D, the finite support map $\mathsf{D} \to \mathcal{P}^+$ is a monad morphism onto the affine-commutative finite structure, reproducing the $\mathcal{P}^+$ growth on cardinalities, so D fails likewise. On the other side, $\mathsf{Maybe} = X + 1$ and $E \times (-)$ are polynomial, so they compose by Proposition 5.2. Both $\mathcal{P}^+$ and D are affine and commutative yet fail; Maybe and the writer are non-affine yet compose. □

5.4. **The categorical converse.** Proposition 5.6 is necessary but does not by itself pin M as a functor: cardinalities cannot separate M from a polynomial functor with the same finite counts. The clean categorical converse we want is

Lemma 5.8 (converse)**.** *If the essential image of* $[\![-]\!]_M$ *is closed under* $\circ$ *— equivalently,* $M \circ N \circ M$ *is of the form* $[\![r]\!] \circ M$ *for every polynomial* N *— then* M *is polynomial.*

We prove Lemma 5.8 for every monad that arises in practice, reducing the sole residual to a single corner we argue is vacuous (Conjecture 5.17). The frame is classical: over **Set**, *M is polynomial $\iff$ M preserves connected limits* (wide pullbacks together with cofiltered limits) $\iff$ M is familially representable (Carboni–Johnstone; [3, §1], [4]). Two features cut the problem down.

First, M is a monad, hence a retract of its square.

Lemma 5.9 (Retract reduction)**.** *For a monad M on* **Set***: M is polynomial $\iff$ $M \circ M$ preserves connected limits.*

Proof. ($\Rightarrow$) Polynomial functors are closed under composition [3], so $M \circ M$ is polynomial and preserves connected limits. ($\Leftarrow$) The right-unit law $\mu \circ \eta M = 1_M$ exhibits M as a *split retract* of $M \circ M$ (section ηM, retraction μ). Limit-preservation passes to retracts — if the comparison map for $M \circ M$ is invertible then so is the one for its retract M, by a diagram chase using naturality of comparison maps in the functor variable. Hence M preserves connected limits, so M is polynomial. $\square$

Second, the trailing M cannot be cancelled, and this is now an exact statement rather than a worry. Closure at $p = q = \mathsf{y}$ yields only the single self-composite identity $M \circ M \cong [\![r]\!] \circ M$, and that identity is categorically inert.

Proposition 5.10 (The single instance is inert)**.** *If $M \circ M \cong P \circ M$ with P polynomial and nonconstant, then for every connected-limit diagram D, $M \circ M$ preserves D $\iff$ M preserves D. Hence the hypothesis $M \circ M \cong [\![r]\!] \circ M$ carries no information about connected-limit preservation of M beyond Lemma 5.9.*

Proof. Naturally isomorphic functors preserve the same limits, so it suffices to treat $P \circ M$. The comparison map for the composite factors as $\varphi_{PM} = \big(P(\lim MD) \cong \lim(P \circ MD)\big) \circ P(\varphi_M)$, the left isomorphism holding because P, being polynomial, preserves the connected limit $\lim(MD)$. A nonconstant polynomial $P = \sum_i (-)^{B_i}$ (some $B_i \neq \emptyset$) reflects isomorphisms, so φ_{PM} is invertible $\iff$ φ_M is. Thus $M \circ M$ preserves D $\iff$ M does. $\square$

This is the precise form of the trailing-M obstruction: precomposition $(-) \circ M$ is not conservative, and the self-composite relates M only to itself, so any proof of Lemma 5.8 must use either the *full* family $\{M \circ N \circ M \cong [\![r_N]\!] \circ M : N \text{ polynomial}\}$ or a genuinely different invariant. We supply the latter, and it settles every monad that occurs.

5.5. **The stabilizer invariant, and the dichotomy.** A natural isomorphism of **Set**-endofunctors is $\mathrm{Aut}(X)$-equivariant at each X (naturality at automorphisms $\sigma\colon X \to X$), so it preserves point-stabilizers.

Lemma 5.11 (Stabilizer invariant)**.** *If $\Theta\colon F \cong G$ then $\mathrm{Stab}_{FX}(x) = \mathrm{Stab}_{GX}(\Theta_X x) \subseteq \mathrm{Aut}(X)$ for all X, x. Hence the family of subgroups $\mathcal{S}(F) := \{\mathrm{Stab}_{FX}(x)\}$, up to conjugacy, is an isomorphism invariant of F.*

For a polynomial $P = \sum_i (-)^{B_i}$ and *any* M, an element of PMX is a pair $(i, f\colon B_i \to MX)$, whose stabilizer is $\bigcap_{b\in B_i} \mathrm{Stab}_{MX}(f(b))$ — an *intersection of MX-element stabilizers.* This is the lever that separates $M \circ M$ from $[\![r]\!] \circ M$ for the analytic monads that evade the cardinality law. The universal such monad is the free commutative monoid.

Theorem 5.12 (The multiset monad does not compose)**.** *Let $\mathbb{M}$ be the finite-multiset (free commutative monoid) monad, $\mathbb{M}X = \{$finite multisets on $X\}$. Then $\mathbb{M} \circ \mathbb{M} \not\cong P \circ \mathbb{M}$ for every polynomial P; in particular $\mathbb{M}$ does not have composition-closed image.*

Proof. A multiset $\nu \in \mathbb{M}X$ is a finitely-supported $\nu\colon X \to \mathbb{N}$, with $\mathrm{Stab}_{\mathbb{M}X}(\nu) = \prod_k \mathrm{Sym}(\nu^{-1}(k))$ a *Young subgroup* — it permutes points only *within* the level-blocks of ν. Intersections of Young subgroups are Young, so by the display above every stabilizer in $P\mathbb{M}X$ is a Young subgroup. But for X containing four distinct points a, b, c, d, the element $m = \{\{a, b\}, \{c, d\}\} \in \mathbb{M}\mathbb{M}X$ has

$$\mathrm{Stab}_{\mathbb{M}\mathbb{M}X}(m) = \big(\mathrm{Sym}\{a, b\}\times\mathrm{Sym}\{c, d\}\big) \rtimes \langle (a\,c)(b\,d)\rangle = S_2 \wr S_2 = D_4, \quad |D_4| = 8,$$

the within-block symmetries *together with the block-swap* $(a\,c)(b\,d)$. D_4 contains $(a\,c)(b\,d)$ but not the within-a-block transposition $(a\,c)$ (which would send m to $\{\{c, b\}, \{a, d\}\} \neq m$), so D_4 is not a Young subgroup: a Young subgroup containing $(a\,c)(b\,d)$ would place a, c in one block and hence also contain $(a\,c)$. (Concretely the Young subgroup orders in S_4 are $\{1, 2, 4, 6, 24\}$, and 8 is not among them.) Thus $D_4 \in \mathcal{S}(\mathbb{M}\mathbb{M})$ but $\mathcal{S}(P\mathbb{M})$ consists of Young subgroups only, and Lemma 5.11 gives $\mathbb{M}\mathbb{M} \not\cong P\mathbb{M}$. □

The block-swap is an irreducibly second-order symmetry that a free (polynomial) outer layer cannot manufacture: a single inner $\mathbb{M}$ supplies only within-block (Young) symmetry, whereas a non-free *outer* layer supplies between-equal-block (wreath) symmetry. In the species calculus this is exactly the distinction between the labelled products in $B \bullet A$ (direct-product stabilizers) and the wreath constituents of the self-substitution $A \bullet A$.

The multiset argument reaches only *bounded* nesting: it pits one wreath $H \wr S_r$ against Young subgroups. For monads whose structures nest to unbounded depth — the free (commutative) magma, free operad-algebra monads, with automorphism towers $S_2 \wr \cdots \wr S_2$ — no fixed stabilizer count suffices, and we replace the combinatorics with one algebraic identity.

5.6. **Cycle-index cancellation, and the necessity trichotomy.** Recall the analytic/species dictionary (Joyal; Bergeron–Labelle–Leroux). An *analytic* functor is one of the form $\tilde{A}(X) = \sum_{n\geq 0} A[n] \times_{S_n} X^n$ for a *species* A (each $A[n]$ a finite S_n-set); its *cycle-index series* is

$$Z_A = \sum_{n\geq 0} \frac{1}{n!} \sum_{\sigma\in S_n} \mathrm{fix}(A[\sigma])\, p_\sigma \in R := \mathbb{Q}[[p_1, p_2, \ldots]], \qquad p_\sigma = \textstyle\prod_k p_k^{c_k(\sigma)}.$$

Three standard facts. **(J1)** $A \mapsto \tilde{A}$ is fully faithful: $\tilde{A} \cong \tilde{B}$ iff $A[n] \cong B[n]$ as S_n-sets for all n. **(F)** $\tilde{A}$ is polynomial iff its species is *flat* (every S_n-action on $A[n]$ free), iff $Z_A \in \mathbb{Q}[[p_1]]$ (only p_1-powers: a non-identity permutation fixing a structure contributes a p_λ with $\lambda \neq (1^n)$). **(C)** if $B[0] = \emptyset$ then species substitution is plethysm on cycle indices, $Z_{A \bullet B} = Z_A \circ Z_B$, where $\circ$ is the unique $\mathbb{Q}$-algebra endomorphism of R with $p_k \circ H = H[p_i \mapsto p_{ki}]$. Writing $a_0 = |A[0]| = |M\emptyset|$ and $a_1 = |A[1]|$, a monad has $a_1 \geq 1$ (the unit $\mathrm{Id} \Rightarrow \tilde{A}$ requires an S_1-fixed point of $A[1]$).

The engine is elementary.

Lemma 5.13 (Plethysm right-cancellation). *Let $H \in R$ have zero constant term and nonzero linear coefficient, $H = a_1 p_1 + (\deg \geq 2)$ with $a_1 \neq 0$. Then $(-) \circ H \colon R \to R$ is injective: $F \circ H = G \circ H \Rightarrow F = G$.*

Proof. $(-) \circ H$ is a $\mathbb{Q}$-algebra endomorphism, so it suffices that $D \circ H = 0 \Rightarrow D = 0$. Since H has bottom degree 1 with $H^{(1)} = a_1 p_1$, plethysm gives $p_\lambda \circ H = a_1^{\ell(\lambda)} p_\lambda + (\deg > |\lambda|)$. If $D \neq 0$, let m be its bottom degree; then $(D \circ H)^{(m)} = \sum_{|\lambda|=m} c_\lambda \, a_1^{\ell(\lambda)} p_\lambda$, and linear independence of $\{p_\lambda\}_{|\lambda|=m}$ together with $a_1 \neq 0$ forces every $c_\lambda = 0$, contradicting minimality. □

The hypothesis $a_1 \neq 0$ is sharp: with $a_1 = 0$ (e.g. $H = p_2 + \frac{1}{2}p_1^2$) plethysm collapses in low degree. Both the injectivity and its failure at $a_1 = 0$ are computationally verified.

Theorem 5.14 (Cancellation converse; $M\emptyset = \emptyset$). *Let $M = \tilde{A}$ be an analytic monad with $A[0] = \emptyset$. If the essential image of $[\![-]\!]_M$ is closed under composition then M is polynomial — with no bound on degree or nesting depth.*

Proof. Closure at $p = q = \mathsf{y}$ gives $M \circ M \cong [\![r]\!] \circ M$ for a polynomial $[\![r]\!]$. As $M \circ M$ is analytic it is finitary, so $[\![r]\!] \circ \tilde{A}$ is finitary; and $\tilde{A}X \supseteq A[1] \times X$ (using $a_1 \geq 1$) is unbounded in $|X|$, so an infinite exponent B_s in $[\![r]\!] = \sum_s (-)^{B_s}$ would make $(-)^{B_s} \circ \tilde{A}$ fail to preserve a filtered colimit — forcing all B_s finite. Thus $[\![r]\!] = \tilde{B}$ for a *flat* species B (fact (F)). Evaluating the isomorphism at $\emptyset$ and using $\tilde{A}\emptyset = A[0] = \emptyset$ gives $B[0] = \tilde{B}\emptyset = \emptyset$. So $A \bullet A$ and $B \bullet A$ are finitary species with $\widetilde{A \bullet A} = \tilde{A} \circ \tilde{A}$ and $\widetilde{B \bullet A} = \tilde{B} \circ \tilde{A}$; the functor isomorphism and (J1) give $A \bullet A \cong B \bullet A$, hence equal cycle indices, and by (C)

$$Z_A \circ Z_A = Z_{A \bullet A} = Z_{B \bullet A} = Z_B \circ Z_A.$$

Here $H := Z_A$ has zero constant term ($a_0 = 0$) and $a_1 \geq 1$, so Lemma 5.13 cancels Z_A on the right: $Z_A = Z_B$. But B is flat, so $Z_B \in \mathbb{Q}[[p_1]]$; hence $Z_A \in \mathbb{Q}[[p_1]]$, i.e. A is flat and M is polynomial (F). □

This is precisely the cancellation of the trailing M that Proposition 5.10 forbade at the level of limit preservation — legitimate here because it operates on the *faithful* cycle-index invariant, which the single self-composite does determine. It reaches the monads no stabilizer count can: the free commutative magma (structure counts $1, 1, 3, 15, 105, \cdots = (2n-3)!!$, $a_0 = 0$, a balanced 2^k-leaf tree with automorphism group $S_2 \wr \cdots \wr S_2$, k factors) has unbounded nesting depth yet falls to Theorem 5.14 in one stroke.

For $M\emptyset \neq \emptyset$ the self-composite is non-finitary and its cycle index diverges; the bounded-degree case is recovered by a sharpening of the stabilizer invariant.

Theorem 5.15 (Support-splitting converse; bounded degree)**.** *Let $M = \tilde{A}$ be a bounded-degree analytic monad ($e_{\max} := \sup_l \deg < \infty$; equivalently $\tilde{A}$ finitary), with any $M\emptyset$. If A is non-flat then $\tilde{A} \circ \tilde{A} \not\cong [\![r]\!] \circ \tilde{A}$ for every polynomial $[\![r]\!]$; hence closure forces M polynomial.*

Proof sketch. For $K \leq \mathrm{Sym}(\Omega)$ the support-splittings ($\mathrm{supp}(K) = \bigsqcup_t C_t$ with $K = \prod_t K_t$, $K_t = \{k \in K : \mathrm{supp}(k) \subseteq C_t\}$) are closed under common refinement, so there is a unique finest one; the multiset of *factor degrees* $\{|C_t|\}$ is a conjugacy invariant (this refines Lemma 5.11). With $[\![r]\!] = \tilde{B}$, B flat and bounded, every constituent of $B \bullet A$ is a Young product of A-stabilizers, whose support-indecomposable factors have degree $\leq e_{\max}$. But a non-flat constituent X^d/H ($H \neq 1$), fed the constant colouring by a maximal-degree constituent $X^{e_{\max}}/H_c$, yields in $A \bullet A$ a wreath $K = (H_c)^d \rtimes H$ in which some $h \in H$ links two size-$e_{\max}$ blocks, producing a support-indecomposable factor of degree $\geq 2e_{\max}$. The invariant separates: $A \bullet A \not\cong B \bullet A$. This subsumes Theorem 5.12 (for $\mathbb{M}$, $e_{\max}$ was there avoided by hand). □

Assembling the three obstructions:

Theorem 5.16 (Necessity, trichotomy)**.** *Let M be a non-polynomial monad on* **Set**. *The essential image of $[\![-]\!]_M$ is not closed under composition whenever M is*

(a) super-polynomially growing — *excluded by the cardinality law (Proposition 5.6: powerset, nonempty powerset, distributions, ultrafilter, continuation, filter); or*
(b) analytic with $M\emptyset = \emptyset$ — *excluded by plethysm right-cancellation (Theorem 5.14), at any degree and any nesting depth: free magmas, free operad-algebra monads; or*
(c) analytic of bounded degree*, any $M\emptyset$ — excluded by the support-splitting invariant (Theorem 5.15): the multiset and symmetric-power monads, every bounded-arity finite-signature monad.*

The only monads not covered are analytic with $M\emptyset$ nonempty-finite, unbounded degree and *unbounded nesting depth.*

Conjecture 5.17 (The uncovered corner is vacuous)**.** For a finitary analytic monad ($A[n]$ finite for all n), $a_0 = |M\emptyset|$ finite and unbounded nesting depth are incompatible: unbounded nesting forces arbitrarily deep balanced iterated structures, and a monad multiplication that builds these from a nonempty constant part substitutes closed terms into the leaves, generating infinitely many closed structures ($a_0 = \infty$) — outside the finitary hypothesis. Hence a finitary analytic monad with $a_0 > 0$ has bounded nesting depth and is covered by case (c); the corner of Theorem 5.16 is empty.

We do not claim Conjecture 5.17 as proved; it converts the last gap from an opaque plethysm question into a concrete tension between a nonempty finite constant part and deep free structure, and it explains why no known monad inhabits the corner — each candidate either drops its constant part to $\emptyset$ (Theorem 5.14) or blows it up to ∞ (non-finitary, outside the hypothesis). Assembling sufficiency with the trichotomy:

Theorem 5.18 (Composition = polynomiality; final form)**.** Sufficiency: *if M is polynomial then the essential image of $[\![-]\!]_M$ is closed under composition, with $p \lhd_M q = p \lhd \lceil M \rceil \lhd q$ (Proposition 5.2).* Necessity: *every non-polynomial monad in classes* (a)–(c) *of Theorem 5.16 — hence every monad that arises — has image* not *closed under composition. Modulo the vacuity of the residual corner (Conjecture 5.17), "$[\![-]\!]_M$ has composition-closed image $\iff$ M is polynomial" is an unconditional biconditional. The earlier reliance on an unproved species-theoretic* Plethysm Lemma *is removed: for $M\emptyset = \emptyset$ the required inequality $A \bullet A \not\cong B \bullet A$ follows directly from right-cancellation in the cycle-index ring, bypassing any identification of a "bad" constituent.*

5.7. **Independence.**

Proof of Corollary 1.4. The writer $E\times(-)$ is polynomial ($E\times(-) = \sum_{e\in E}(-)^1$), so it composes by Proposition 5.2; it is not codense ($\Phi(1) = |E|^{|E|} \neq |E|$ for $|E| \geq 2$), so it is not full by Theorem 1.2. The distribution monad D is affine and codense (Example 4.3), so it is full by Theorem 1.2; it is not polynomial (it does not preserve wide pullbacks — joints are not determined by marginals), so it fails composition by Corollary 5.7. Codensity and polynomiality are therefore logically independent properties of M. □

6. Conclusion

For a monad M on **Set**, the effectful container extension $[\![-]\!]_M = [\![-]\!] \circ M$ names endofunctors faithfully always, names them *fully* exactly when M is codense, and has an image closed under composition when M is polynomial. These are two different Kan-theoretic invariants of the effect monad, and across the standard effects they pull in opposite directions: probability is faithful but not composable, error is composable but not faithful. The folklore "composes iff commutative and affine" is simply wrong — D is affine and commutative but does not compose, Maybe is neither yet does — and the true obstruction to composition is a representability dichotomy, not a cohomology class or a distributive-law existence question.

For the compositional modelling of effectful agents this draws a sharp line. An agent that "may decline" (Maybe-positions) chains cleanly into other agents but cannot always be recovered from its behaviour; an agent that "answers probabilistically" (D-positions) is recoverable from its behaviour but does not chain into a single probabilistic agent. One pays for declining in faithfulness and for probability in composability, and the invariant that sets each price is now explicit.

Three questions remain, each precisely located.

(1) **The composition converse (the vacuous corner, Conjecture 5.17).** Does closure of the essential image under $\circ$ force M to be polynomial? Yes for every monad that arises (Theorem 5.16): the cardinality law excludes the super-polynomial monads, plethysm right-cancellation (Lemma 5.13) excludes every analytic monad with $M\emptyset = \emptyset$ at any nesting depth, and the support-splitting invariant excludes every bounded-degree analytic monad. The trailing-M obstruction was sharpened into the exact "no information" statement of Proposition 5.10 at the level of limit preservation, then circumvented by cancelling on the finer cycle-index invariant. The one class left uncovered — analytic, $M\emptyset$ nonempty-finite, unbounded degree and nesting depth — is argued vacuous in Conjecture 5.17; proving that vacuity would make Theorem 5.18 an unconditional biconditional with no side hypothesis.

(2) **Rigidity of D (Lemma 4.4).** Full faithfulness of $[\![[\!]_M\mathsf{D}]-$ is conditional on $\mathrm{Nat}(\mathsf{D}, \mathsf{D}) = \{\mathrm{id}\}$; the reduction of Example 4.3 is unconditional. A clean proof of this rigidity fact would remove the one assumption in Theorem 1.2's probabilistic case.

(3) **Kleisli-bifunctoriality (Remark 5.5).** The composition product is defined on objects and is strong semigroupal on pure morphisms; it upgrades to the full Kleisli category exactly when M is codense. Thus the two theorems meet: composition needs polynomiality to exist and codensity to become bifunctorial on effectful morphisms. The monads for which both hold — the affine polynomial monads — are worth isolating.

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